

Refinery Safety Standard Operating Procedure (SOP-HSE-042)

Hazardous Fluid Containment & Emergency Isolation Protocol • Safety First

1. Objective & Regulatory Compliance

Standard Operating Procedure: Refinery Hazardous Hydrocarbon Isolation and Emergency Shutdown (ESD-01).

Applies to all operating units handling Class 1 flammable gases, toxic hydrogen sulfide (H2S), and high-pressure steam.

Compliance standard: OISD-STD-105 (Work Permit System) and Petroleum Rules.

2. Mandatory Personal Protective Equipment (PPE)

Level 1 Areas: Flame-retardant anti-static overalls, safety helmet, steel-toed boots, impact goggles, H2S personal multi-gas detector.

Level 2 (Active Breaching of Containment): Air-purifying escape respirator or positive-pressure SCBA unit.

No personal electronic devices, non-intrinsically safe radios, or portable laptops permitted inside battery limit without hot work permit.

3. Emergency Shutdown (ESD) Initiation Protocol

Immediate ESD push-button activation is MANDATORY upon detection of:

- Flammable gas concentration exceeding 20% LEL (Lower Explosive Limit) at unit perimeter.
- Major hydrocarbon leak > 10 kg/min with potential for ignition.
- Compressor unbalance resulting in vibration exceeding 8.0 mm/s RMS.
- Loss of cooling water circulation or instrument air header pressure dropping below 3.5 bar.

4. Autonomous AI & Digital Safety Guardrail

CRITICAL SAFETY RULE: Under no circumstances shall an AI system, automated agent, or machine learning algorithm be connected to trigger, suppress, or modify Emergency Shutdown (ESD) interlocks or safety relief valves.

All recommendations generated by AI workbenches are advisory decision-support tools only.

Physical human authorization by the Shift Superintendent is strictly required before any field equipment actuation.